

A dynamic feature selection technique to detect DDoS attack

Usham Sanjota Chanu^{*}, Khundrakpam Johnson Singh, Yambem Jina Chanu

Department of CSE, National Institute of Technology, Manipur, India

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ABSTRACT

Distributed Denial of Service (DDoS) attacks are a weapon of choice for hackers that are accountable for degradation in network performance, website downtime, and sometimes crashing of servers, etc. Because the feature selection mechanism is still naive, detecting DDoS attacks is difficult. There is a need to efficiently select features from the attack dataset, which also needs to be further analyzed. Simply placing the features as input to the classification model will result in an irrelevant class with poor performance. In the paper, in order to select the effective features, we proposed a voting-based hybrid feature selection technique that was cross-analyzed by three correlation methods. The hybrid method not only reduces the dimensions of the features and removes redundancy among them, but also provides the best relevant features for classification. The deployment of a multilayer perceptron with genetic algorithm (MLP-GA) as a classifier outperforms the conventional classifiers by furnishing an accuracy of 98.8%, a false positive rate of 0.6%, and the capability of early detection. The proposed method has been comprehensively validated using various publicly available benchmark datasets.

1. Introduction

Aside from land, air, and water space, cyber warfare appears to be a dominant mode of modern warfare. The exponential growth of Internet interconnection with less security provides a favourable environment for cyberattacks. A cyberattack is an assault launched by cyber criminals that leads to disastrous and grievous consequences. Cybercriminals maliciously infect computers, steal information, demand ransom, or use a breached computer as a launchpad for various attacks. Cybercriminals use a variety of techniques to launch cyberattacks, which include ransomware, malware attacks, phishing, denial of service (DoS), etc.

A DoS attack overloads the targeted system's capacity, rendering its resources inaccessible to its intended users [1]. The DoS attack sends a massive flood of requests to the targeted server, causing the server to crash or become unavailable. The common types of DoS attacks include the SYN Flood attack [2], ICMP Flood attack [3], etc. The DoS attack has now evolved into a sophisticated attack known as the Distributed Denial of Service (DDoS) attack. A DDoS attack is a critical version of DoS where multiple systems in different geographical areas collaborate to carry out DoS attacks on a targeted server [4]. As a result, the consequences will be much more adverse than those of a DoS attack. An example of a DDoS attack is shown in Fig. 1, where the attacker controls the compromised nodes located in different geographical areas through a Command & Control center. The attacker sends illegal commands to

these nodes to target the victim's server.

The main motives for DDoS attacks are to carry out extortions, to target websites that have ideological conflicts, business squabbles, political gains, and sometimes for fun [4]. In a recent market survey, it was found that DDoS attacks are one of the major forms of cyber warfare that cause huge concerns. We need to know more about DDoS attacks and effective detection mechanisms, as we rely on the Internet for the majority of our activities.

The detection of DDoS attacks relies on the effectiveness of the feature selection from the attack dataset and the deployment of a classification algorithm that has high performance. The survey conducted for most research papers reveals that the feature selection technique is lacking in the majority of them. Papers, in [5] manually considered four features, 16 features in [7], 3 features in [8], 16 features in [9], 6 features in [13] without even checking their effectiveness in the detection. Some of the research papers simply reduce the dataset dimensions, as in [10], and select features using conventional feature extraction techniques.

Most of the research work prioritises their work on the classification model rather than feature selection; there was no discussion on how the features are selected in [11,13]. Some of the researchers in [12] deploy anomaly-based detection by tagging the legitimate packets and thereby restricting the untagged packets from accessing the server. This technique will increase the false-positive rate when legitimate traffic is

^{*} Corresponding author.

E-mail address: chanu06atcs012@gmail.com (U.S. Chanu).

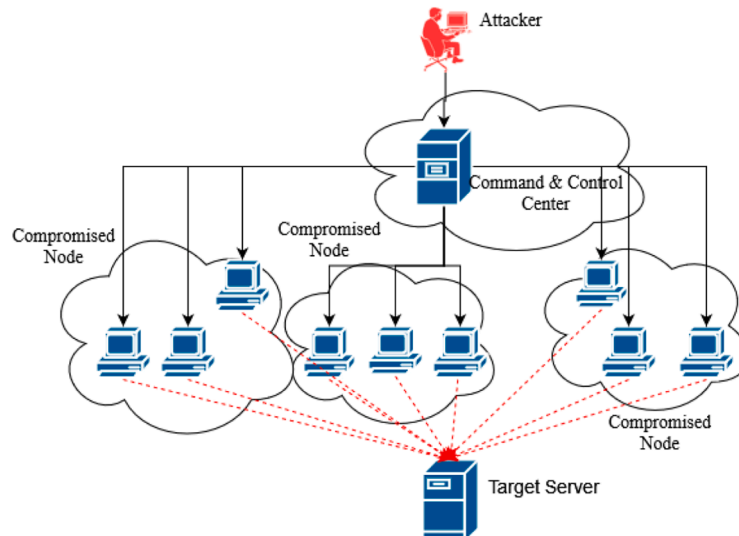


Fig 1. An example of DDoS attack.

wrongly tagged. Research in [14] studied only the flooding or volumetric type of DDoS attack generated by automated tools, thereby not exploring generalised DDoS attacks. There is a huge lack of research in the selection of features and deployment of techniques that are cross-checked to find out whether any irrelevant or duplicate features are still present before being fed to the classification algorithm. Taking into account only one or a few features not only results in unrelated class categories but also increases the false positive rate. Also, the selection of too many unrelated features not only degrades performance but also increases the computation overhead.

Taking into account the gap in the existing research, we proposed a hybrid feature selection method based on a three-step wise filter to systematically select only the effective features with no duplication and the deployment of a multilayer perceptron with a genetic algorithm (MLP-GA) as a classifier. GA is a search-based algorithm based on the concepts of genetics and natural selection. The feature selection methods and the MLP-GA algorithm are explained in Sections 3.2 and 3.5, respectively. We can reduce the dimension of the dataset using this hybrid method by selecting only the highest priority features that are correlated to one another. In the paper, we examined a generalised DDoS attack and ensured to make the process more accurate with less computation overhead by selecting the most critical features and eliminating the redundant and irrelevant ones.

In feature selection, it is best practice to select the correlated features and ignore those that are less correlated. The values of the features present in the benchmark dataset might be continuous or ordinal. As a result, the deployment of a single correlation mechanism is not sufficient to draw a conclusion. Therefore, we deployed three correlation mechanisms: Pearson, Spearman, and Kendall, to check the relationship between the features and minimize the error generated by any of them. Predicting without any prior correlation analysis may be inefficient, less reliable, and more uncertain. Only those features that have been selected and correlated are fed to the MLP-GA in order to classify the class category as an attack or benign class. The main contributions in the paper are listed below:

- This paper proposes a hybrid voting based ensemble feature selection mechanism with subsequent correlation techniques to pick the best features from the dataset with no redundancy.
- We analyze and compare various activation functions in a MLP classification model to retrieve the best fitting function.
- We introduced GA with MLP to achieve an overall improvement in classification performance.

The rest of the paper is organized as follows: Section 2 provides the related works and outlines the various methodologies used earlier. Section 3 describes the proposed methods that include data pre-processing, feature selection, and the MLP-GA algorithm. Section 4 provides the results and discussions and is finally concluded in Section 5.

2. Related works

Dong and Sarem [5] approach was to analyze the traffic behavior of DDoS attacks on the Software Defined Network (SDN) with two methods of DDoS detection. Only four features, such as flow length, flow duration, flow size, and flow ratio, are examined for the detection purpose. The two methods of detection include the utilization of the degree of attack to identify DDoS attacks and the deployment of K-Nearest Neighbors (KNN) algorithm based on Machine Learning (ML) to discover the DDoS attack.

Poongodi et al. [6] focused on a safe routing mechanism based on the node trust matrix in Vehicular Ad-hoc networks (VANET) to isolate and detect DDoS attacks. Based on trust components such as frequency value statistics, trust hypothesis statistics, residual energy, trust policy, and data factor, they presented a trust matrix to develop a secure communication network to avoid trespassers into the system. The major components of evaluating the trust matrix were trust policy, residual energy, frequency value statistics, and data factor.

Erhan and Anarim [7] proposed a Matching Pursuit (MP) based on low-density DDoS attack detection using multiple characteristics of network traffic. The CAIDA and Boğaziçi University DDoS attack dataset (BOUN) are evaluated using the K-mean Singular Value Decomposition (K-SVD) algorithm. They deployed a hybrid DDoS detection framework, which is a combination of anomaly and misuse detection methods. The hybrid technique deployed is an integration of the Matching Pursuit mean projection method, wavelength techniques, and a decision-making mechanism using an Artificial Neural Network (ANN).

Benamar et al. [8] proposed an anomaly detection mechanism that uses the continuous ranked probability score (CRPS) statistical mechanism and an exponentially smoothing (ES) scheme to enable DoS and DDoS attack detection. The normal data are used as input to the CRPS-ES metric. The CRPS-ES computes a non-parametric decision threshold using kernel density estimation. Attacks are detected when the CRPS-ES measurement between the test and normal traffic exceeds the threshold value computed.

Nisha et al. [9] created an SDN dataset and proposed a hybrid machine learning model to classify the benign traffic from the DDoS attack traffic. The hybrid model is comprised of Support Vector classifier (SVC)

Table 1
CICDDoS 2017 Dataset features.

Feature no.	Feature label	Feature no.	Feature label	Feature no.	Feature label
1.	Destination Port	28.	Bwd IAT Std	54.	AvgFwd Segment Size
2.	Flow Duration	29.	Bwd IAT Max	55.	AvgBwd Segment Size
3.	Total Fwd Packets	30.	Bwd IAT Min	56.	Fwd Header Length
4.	Total Backward Packets	31.	Fwd PSH Flags	57.	FwdAvg Bytes/Bulk
5.	Total Length of Fwd Packets	32.	Bwd PSH Flags	58.	FwdAvg Packets/Bulk
6.	Total Length of Bwd Packets	33.	Fwd URG Flags	59.	FwdAvg Bulk Rate
7.	Fwd Packet Length Max	34.	Bwd URG Flags	60.	BwdAvg Bytes/Bulk
8.	Fwd Packet Length Min	35.	Fwd Header Len	61.	BwdAvg Packets/Bulk
9.	Fwd Packet Length Mean	35.	Bwd Header Length	62.	BwdAvg Bulk Rate
10.	Fwd Packet Length Std	37.	Fwd Packets/s	63.	SubflowFwd Packets
11.	Bwd Packet Length Max	38.	Bwd Packets/s	64.	SubflowFwd Bytes
12.	Bwd Packet Length Min	39.	Min Packet Length	65.	SubflowBwd Packets
13.	Bwd Packet Length Mean	40.	Max Packet Length	66.	SubflowBwd Bytes
14.	Bwd Packet Length Std	41.	Packet Length Mean	67.	Init_Win_bytes_forward
15.	Flow Bytes/s	42.	Packet Length Std	68.	Init_Win_bytes_backward
16.	Flow Packets/s	43.	Packet Length Variance	69.	act_data_pkt_fwd
17.	Flow IAT Mean	44.	FIN Flag Count	70.	min_seg_size_forward
18.	Flow IAT Std	45.	SYN Flag Count	71.	Active Mean
19.	Flow IAT Max	46.	RST Flag Count	72.	Active Std
20.	Flow IAT Min	47.	PSH Flag Count	73.	Active Max
21.	Fwd IAT Total	48.	ACK Flag Count	74.	Active Min
22.	Fwd IAT Mean	49.	URG Flag Count	75.	Idle Mean
23.	Fwd IAT Std	50.	CWE Flag Count	76.	Idle Std
24.	Fwd IAT Max	51.	ECE Flag Count	77.	Idle Max
25.	Fwd IAT Min	52.	Down/Up Ratio	78.	Idle Min
26.	Bwd IAT Total	53.	Average Packet Size		
27.	Bwd IAT Mean				

and Random Forest (RF) models that classify the traffic using SVC results, which are then filtered again by RF. The proposed SVC-RF model exhibits the highest accuracy of 98.8%.

Wei et al. [10] proposed a hybrid model combining two deep learning models, an Autoencoder (AE) and an MLP. The AE, which is an unsupervised feed-forward neural network, is used for feature extraction. The purpose of AE is to reconstruct the input from the output and to perform dimensionality reduction. The extracted features are then fed into the MLP model as inputs and are now used to train the MLP model as a classifier to detect different DDoS attack types. Relu and softmax are the activation functions deployed in the hidden layer and output layer of the MLP, respectively. The model used the CICDoS 2019 dataset, a commonly used dataset for DDoS attack classification and detection.

Saied et al. proposed an anomaly-based supervised ANN classification model [11] for the detection of DDoS attacks with a sigmoid activation function. The statistical features such as instances of packet headers, including source addresses, IDs, and sequence numbers, along with source and destination port numbers, are considered the input pattern for the model. The method can detect DDoS attacks that are carried out using Transmission Control Protocol (TCP), User Datagram

Protocol (UDP), and Internet Control Message Protocol (ICMP). This is based on the characteristics of the traffic pattern that distinguish between the attack and legitimate traffic. These patterns are input as a training dataset for the ANN. The method requires continuous observation of the traffic flow. As a result, DDoS detectors are installed on multiple networks. Anomalies are detected when the number of packets transmitted is higher than the pre-defined threshold value.

Karrer et al. proposed an Edge-based Capabilities (EC) [12] that takes the advantages of their framework by combining end-to-end authentication with network-based control. The EC provides a technique that cryptographically tags legitimate traffic. This tag is generated by senders in EC when they receive permission to send from the intended receiver. Untagged and wrongly tagged traffic are treated as attack traffic. The EC applications known as gates that reside at the network edge inspect for untagged or wrongly tagged packets and provide priority to the legitimate tagged packets.

Oke et al. proposed a technique that utilizes a combination of a Bayesian classifier and a Random Neural Network (RNN) [13] that can discriminate between an attack and normal traffic during a DDoS attack. The input is the statistical features that include bit rate, increase in bit

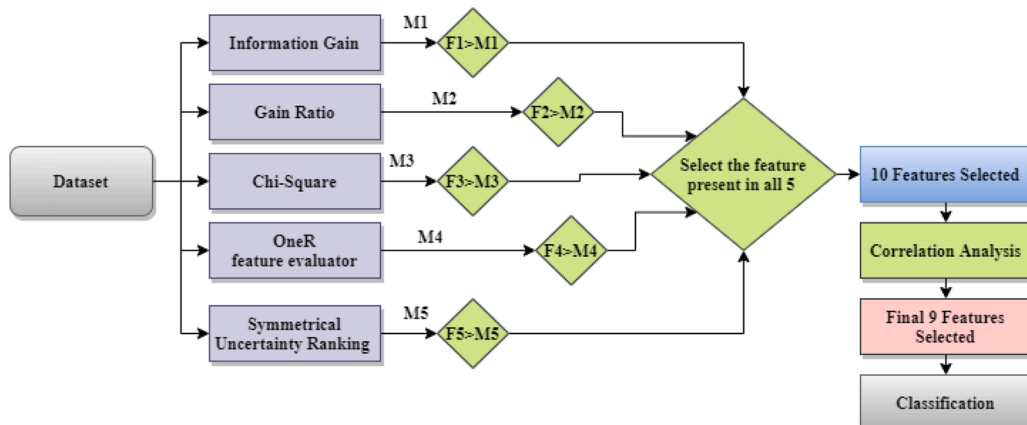


Fig. 2. Hybrid feature selection model.

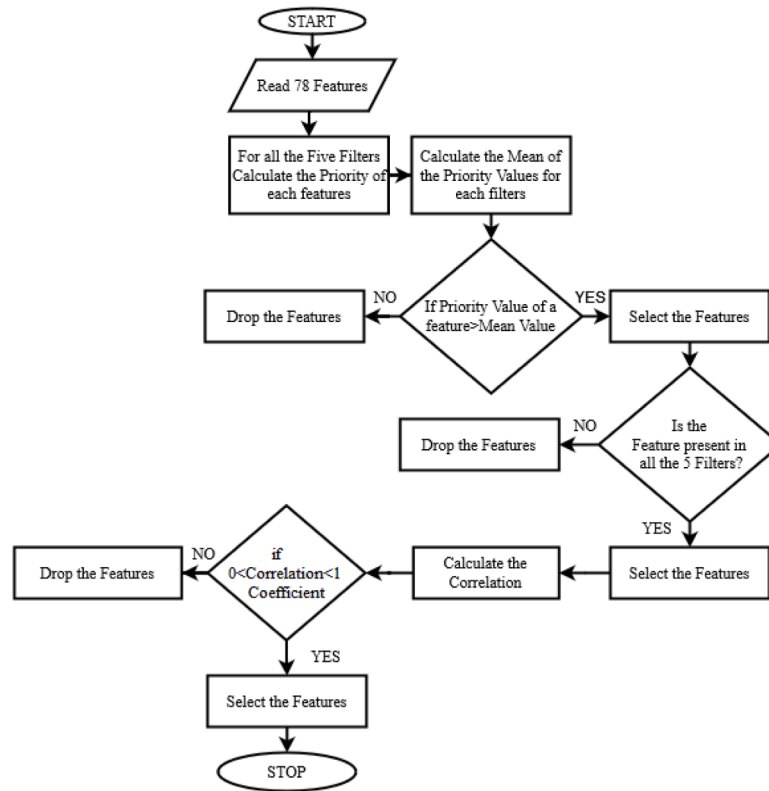


Fig 3. Flowchart for the features selection.

rate, entropy, the hurst parameter, delay, and delay rate of the incoming traffic. Once the features were selected, the estimates of probability density functions as histograms for each feature were calculated, and likelihood ratios for each of the features were computed. These ratios can be interpreted as first-level decisions. The method is validated using the DARPA 98 to 2000 datasets.

Siaterlis et al. proposed an MLP-based method to detect UDP flooding attacks [14]. They considered the network state as input to the MLP. The MLP classify the incoming traffic into either two classes consisting of attack and normal or three classes consisting of source, victim, and normal. The network state is defined by five factors, consisting of the UDP ratio, ICMP ratio, flow length distribution, flow lifetime distribution, and flow generation rate. The first two factors are the result of packet capture, and the last three are the result of Netflow collector. They illustrated that addition of more neurons to the hidden layer does not imply any enhancement of classification results. The method is tested in the dataset generated by the Tribe Flood Network 2000 (TFN2K) tool, which is capable of launching UDP floods.

3. Proposed method

3.1. Data pre-processing

Data preprocessing is performed to clean the data, normalize the data, and select a subset of features. This step is crucial because the noise present in the data could degrade performance. Data cleaning is required because real captured data can have many irrelevant and missing values. In the paper, we utilized CICDDoS 2017 dataset [15] to carry out the experiments. CICDDoS 2017 Dataset features are shown in Table 1.

3.2. Feature selection

In the paper, we proposed a hybrid feature selection method based on a three-step wise filter, as shown in Fig. 2. The graphical representation of the movement of data in each step of the filter to yield the final set of features is shown by the flowchart in Fig. 3 and illustrated at Algorithm 1. In the first step, we deploy five feature selection algorithms, namely information gain [16], gain ratio [17], chi-square [18], OneR feature evaluator [19], and symmetrical ranking [20], to calculate the priority of each feature in the database.

We are interested in these filters because of their high computing

Table 2
Calculated average rank value and the subsequent feature selected.

Filter Deployed	Mean value	Feature serial number above mean value	Common Features
Total features present in the dataset is 78			
Information Gain	M1=0.21781	1,58,51,5,67,39,7,33,41,40,59,9,52,24,38,21,2,35,19,5,64,22,10,35,13,53,16,14,11	2,5,19,21,
Gain Ratio	M2= 0.0867	1,58,47,14,45,21,46,38,65,57,6,41,40,62,64,2,5,35,24,37,59,67,51,50,63,10,39,61,53,13,52,19,33,66,68	24,35,41, 53,64,67
Chi-Square	M3= 0.16113	1,46,45,47,53,13,11,14,24,19,21,2,35,64,23,61,67,69,38,40,22,41,50,17,39,51,52,5,27,58	
OneR Feature Evaluator	M4=0.3731	58,51,64,5,33,39,40,41,7,21,19,24,2,35,53,9,22,67,16,15,17,21,46,18,45,38,54,3,6,57,36,8	
Symmetrical Uncertainty Ranking	M5=0.2596	58,33,1,51,7,5,55,39,64,24,40,41,57,6,21,53,13,35,9,52,22,23,19,18,11,67,2,34,35,17,16,15,36,10,14,3,54,25	

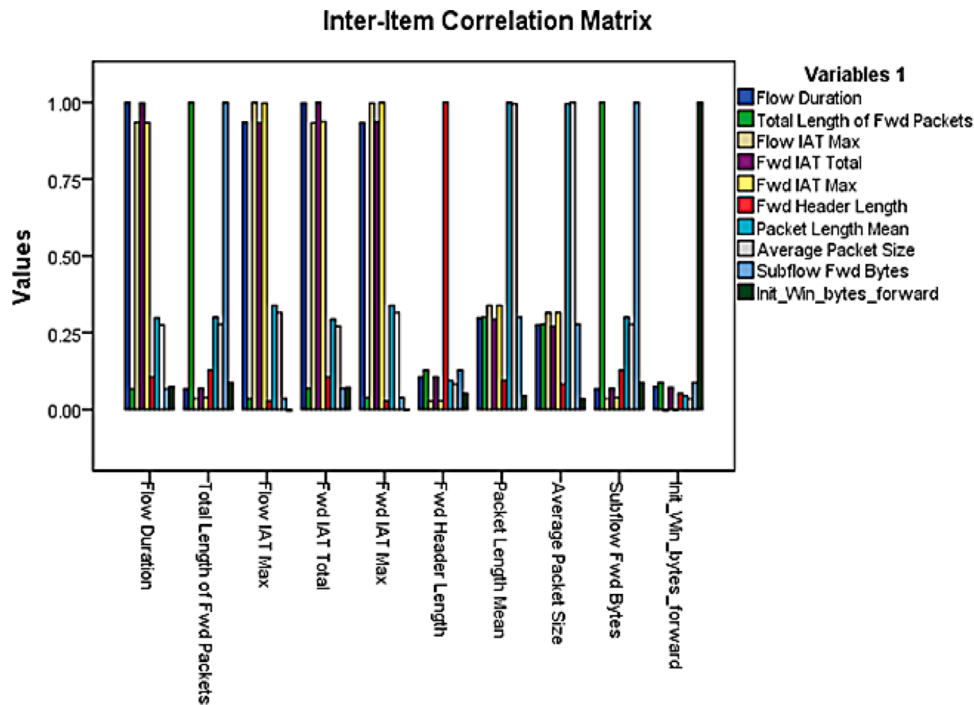


Fig. 4. Ten features selected after the second filtering step.

speed, scalability to high-dimensional datasets, and low overfitting. When they are viewed individually, their limitations arise due to their nature of not depending on a classifier. They ignore feature dependencies and look at each feature separately.

For each ranking algorithm or filter, we compute the average rank (M1, M2, M3, M4, M5) and select those sets of features (F1, F2, F3, F4, F5) whose value is above the mean priority value. The calculated value for all the five ranking features as well as the number of features whose priority value is above their respective means are shown in Table 2. Once we have the selected features for each feature selection algorithm, we now shift to the second step of feature selection.

In the second step, we consider only those features that are present in all the feature selection algorithms as shown in Table 2. After the end of the second step, only 10 features out of 78 are left, as shown in Fig. 4. In the next step, we need to analyze whether these 10 features are correlated with one another or if redundant features are present. In the third step, we have to consider only those features out of the 10 that are correlated to one another. Selecting features that are correlated with one another and without redundancy will result in better classification performance.

3.2.1. Correlation analysis

In the third step of filtering to select the most efficient features, we deploy a correlation to determine whether or not there are relationships between two features. Three cases can be drawn based on the value of the correlation obtained.

Case 1: A correlation coefficient nearing 0, but either positive or negative, indicates minimal or no relationship between the two features.

Case 2: A correlation coefficient near +1 but not equal to +1 indicates that the relationship between the two features is positive. This implies the existence of a strong relationship between the features considered. However, when we have a correlation coefficient value equal to +1 between two features, this indicates that the two features are highly similar, and hence we drop one of the features.

Case 3: A correlation coefficient nearing -1 indicates that the relationship between the two features is negative. This implies a weak relationship between the features considered.

In the paper, we examined three different correlations to determine the significance of the features selected by the ensemble feature selection algorithms. We choose multiple correlations to verify whether the selected features show important significance in each of the different correlation algorithms.

Pearson correlation [21]: It is used to measure the degree of the

Table 3
Pearson correlation.

Pearson Correlation N = 20,000									
	Flow Duration	Fwd Packet Length Mean	Max Packet Length	Packet Length Mean	Packet Length Std	Packet Length Variance	Average Packet Size	AvgFwd Segment Size	Init_Win_bytes_forward
Flow Duration	1	.093	.354	.299	.311	.186	.275	.093	.075
Fwd Packet Length Mean	.093	1	.361	.446	.370	.285	.447	1.00	.066
Max Packet Length	.354	.361	1	.914	.983	.917	.890	.361	.051
Packet Length Mean	.299	.446	.914	1	.951	.828	.995	.446	.045
Packet Length Std	.311	.370	.983	.951	1	.936	.934	.370	.051
Packet Length Variance	.186	.285	.917	.828	.936	1	.815	.285	.003
Average Packet Size	.275	.447	.890	.995	.934	.815	1	.447	.035
AvgFwd Segment Size	.093	1.00	.361	.446	.370	.285	.447	1	.066
Init_Win_bytes_forward	.075	.066	.051	.045	.051	.003	.035	.066	1

Table 4
Kendall Correlation.

KendallCorrelation Coefficient $N = 20,000$									
	Flow Duration	Fwd Packet Length Mean	Max Packet Length	Packet Length Mean	Packet Length Std	Packet Length Variance	Average Packet Size	AvgFwd Segment Size	Init_Win_bytes_forward
Flow Duration	1.00	.307	.320	.302	.278	.278	.271	.307	.125
Fwd Packet Length Mean	.307	1.00	.804	.836	.612	.612	.825	1.00	.004
Max Packet Length	.320	.804	1.00	.907	.854	.854	.893	.804	.069
Packet Length Mean	.302	.836	.907	1.00	.717	.717	.977	.836	.029
Packet Length Std	.278	.612	.854	.717	1.00	1.00	.706	.612	.149
Packet Length Variance	.278	.612	.854	.717	1.00	1.00	.706	.612	.149
Average Packet Size	.271	.825	.893	.977	.706	.706	1.00	.825	.031
AvgFwd Segment Size	.307	1.00	.804	.836	.612	.612	.825	1.00	.004
Init_Win_bytes_forward	.125	.004	.069	.029	.149	.149	.031	.004	1.00

Table 5
Spearman Correlation.

Spearman Correlations Coefficient									
	Flow Duration	Fwd Packet Length Mean	Max Packet Length	Packet Length Mean	Packet Length Std	Packet Length Variance	Average Packet Size	AvgFwd Segment Size	Init_Win_bytes_forward
Flow Duration	1.00	.424	.443	.428	.385	.385	.391	.424	.175
Fwd Packet Length Mean	.424	1.00	.919	.940	.760	.760	.937	1.00	.017
Max Packet Length	.443	.919	1.00	.973	.901	.901	.970	.919	.106
Packet Length Mean	.428	.940	.973	1.00	.812	.812	.996	.940	.053
Packet Length Std	.385	.760	.901	.812	1.00	1.00	.810	.760	.200
Packet Length Variance	.385	.760	.901	.812	1.00	1.00	.810	.760	.200
Average Packet Size	.391	.937	.970	.996	.810	.810	1.00	.937	.054
AvgFwd Segment Size	.424	1.00	.919	.940	.760	.760	.937	1.00	.017
Init_Win_bytes_forward	.175	.017	.106	.053	.200	.200	.054	.017	1.00

relationship between linearly related features. The Eq. (1) is used to calculate the Pearson r correlation. Table 3 shows the calculated correlated values using Pearson correlation. In this table, we remove the features having negative values and retain only those having positive values.

$$P_{r(x,y)} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}} \quad (1)$$

where $P_{r(x,y)}$ = Pearson correlation coefficient between x and y

n = number of observations, x_i = value of x (for i th observation), y_i = value of y (for i th observation).

Kendall rank correlation [22]: It is a non-parametric test that measures the dependency between two features. Eq. (2) is used to calculate the value of the Kendall rank correlation. Table 4 shows the calculated correlated values using Kendall rank correlation. Similar to the Pearson correlation table, we remove features with negative values and keep only those with positive values.

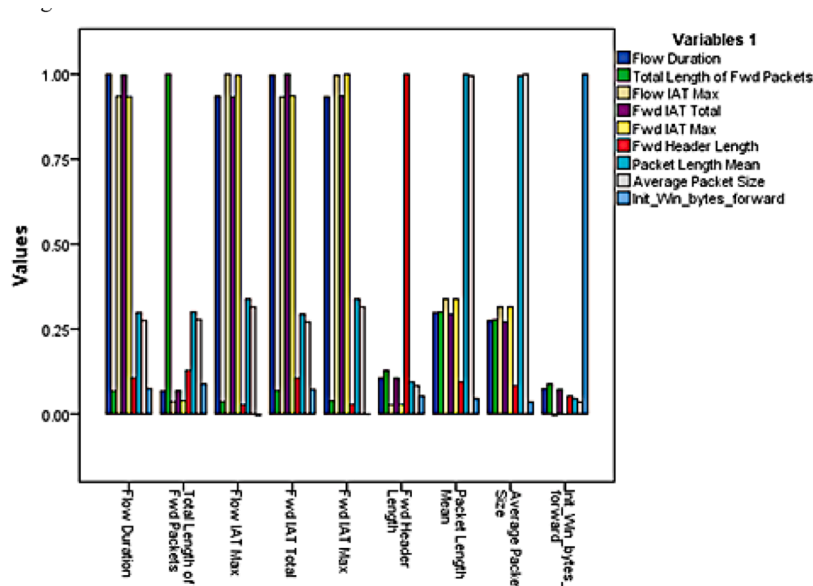
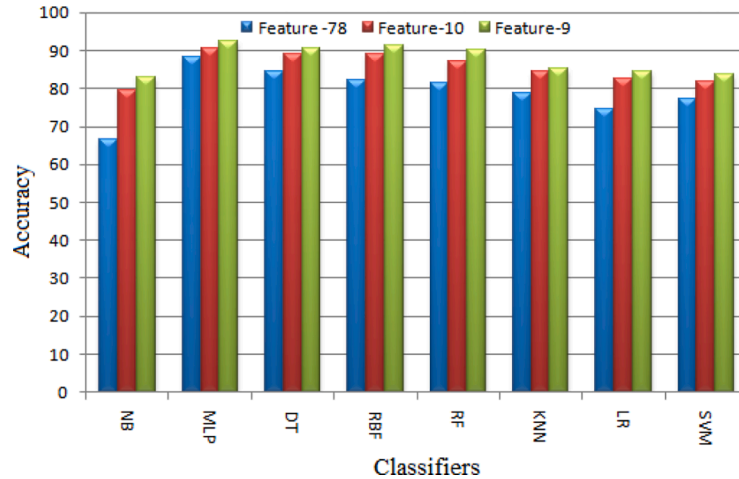


Fig 5. Correlation of the final nine selected features.

Table 6

Performance comparison of conventional classifiers using different number of features.

Classifier	78 Features			10 Features			9 Features		
	Accuracy	Time taken (secs)	False Positive Rate	Accuracy	Time taken (secs)	False Positive Rate	Accuracy	Time taken (secs)	False Positive Rate
Naïve Bayes (NB)	66.4	13.07	31.3	79.4	0.11	19.6	82.8	0.08	17.3
MLP	88.44	32.3	10.6	90.4	2.33	9.2	92.5	1.54	8.3
Decision Tree (DT)	84.4	1043.6	15.8	89	22.19	10.8	90.4	22.15	9.3
RBF Network	82.3	21.8	17.3	88.9	8.3	10.8	91.2	7.2	8.9
Random Forest (RF)	81.3	19.16	18.4	87.2	7.3	12.6	90.1	6.44	9.0
K-Nearest Neighbour (KNN)	78.9	20.2	20.3	84.5	10.6	15.3	85.4	8.4	14.2
Logistic Regression (LR)	74.4	21.6	24.3	82.6	3.33	17.9	84.4	3.16	15.9
SVM	77.2	19.2	21.7	81.8	7.8	18.9	83.8	7.8	16.4

**Fig 6.** Comparison of accuracy for different classifiers across multiple features.

$$Kr(x, y) = \frac{N_x - N_y}{\frac{1}{2}N(N-1)} \quad (2)$$

where, N_x =number of concordant, N_y = Number of discordant.

Spearman rank correlation [23]: It is a non-parametric test that is used to measure the degree of association between two features. The Eq. (3) is used to calculate the Spearman rank correlation. Table 5 shows the calculated correlated values using Spearman rank correlation; in this table too, we remove the features having negative values and retain only those having positive values.

$$Sr = 1 - \frac{6 \sum d_i^2}{N^3 - N} \quad (3)$$

where,

Sr = Spearman rank correlation

d_i = the difference between the ranks of corresponding features

N = number of observations

When we analyze the multiple correlation methods in Tables 3, 4 and 5, we conclude that nine features out of the ten selected features in step two have a positive correlation value. Since the feature, SubflowFwd Bytes, has a negative correlation and hence removed. The classification model's final set of features to be used as input will only have nine features. The correlation of the final nine features is shown in Fig. 5. In the paper, we consider an attack class and a benign class as the two target classes.

3.3. Background

In the paper, we now input the selected features to some of the

conventional classification models, such as Naïve Bayes [24], MLP [25], Decision Tree [26], RBF Network [27], Random Forest [28], K-nearest neighbor (KNN) [5], Logistic Regression (LR) [29], and Support Vector Machine (SVM) [30], and observe the performance metrics generated by each of them. The first set of features (78-Features) are the original features present in the dataset. The second set (10-Features) and third set (9-Features) are those features selected in the second step and third step of feature selection, respectively.

The classification models were supplied with three sets of features, as shown in Table 6, to compare their performance before and after the hybrid feature selection technique was implemented. It is observed from Table 6 that for Feature-78, the accuracy is very low, with a high false positive rate and much time consumption. For Feature-10, there is an increase in accuracy with a lower false positive rate and decreased time consumption. However, with Feature-9, there is a drastic improvement in accuracy with a minimal false-positive rate and less time for computation. The comparison of the various classification models in terms of accuracy for three different feature sets is shown in Fig. 6.

It is observed from Table 6 and Fig. 6 that MLP outperforms the other conventional classification models in various performance metrics. However, there is a need to further analyze the MLP classifier to improve its performance.

3.4. Function of activation function in neural network

Another form of artificial neural network (ANN) is the MLP. MLP is a feedforward artificial neural network model that transform input data sets to a set of appropriate outputs. A minimum of three layers of nodes fully connected to the following ones makes up the simplest MLP: an input layer, a hidden layer, and an output layer. Except for the nodes in the input layer, the nodes of the layers are neurons with nonlinear

Table 7

Summary of performance with variation in activation function.

Model	Activation Function		Sum of Square errors		Training Time (Seconds)	Classification	
	Hidden Layer	Output Layer	Training	Testing		Attack	Benign
Deep Neural Network	Hyperbolic Tangent	Identity	2095.23	927.31	05.20	84%	88%
	Sigmoid	Identity	2157.29	909.05	03.32	85%	89%
	Hyperbolic Tangent	Softmax	6690.11	2925.68	03.46	88%	90%
	Hyperbolic Tangent	Hyperbolic Tangent	2257.18	955.79	02.80	89%	90%
	Hyperbolic Tangent	Sigmoid	2161.79	914.37	02.74	89%	91%
	Sigmoid	Softmax	6525.81	2826.96	03.21	89%	91%
	Sigmoid	Hyperbolic Tangent	2159.05	954.37	02.42	88%	90%
	Sigmoid	Sigmoid	2017.18	914.15	03.59	87%	88%
MLP	Hyperbolic Tangent	Identity	2061.62	889.92	01.45	89%	91%
	Sigmoid	Identity	2214.39	938.32	01.35	88%	90%
	Hyperbolic Tangent	Sigmoid	6043.03	2562.54	01.36	91%	94%
	Hyperbolic Tangent	Hyperbolic Tangent	2363.15	1045.15	01.37	90%	93%
	Hyperbolic Tangent	Softmax	2129.42	910.95	01.36	91%	92%
	Sigmoid	Softmax	7434.70	3144.78	01.35	89%	93%
	Sigmoid	Hyperbolic Tangent	2236.77	967.25	01.36	89%	92%
	Sigmoid	Sigmoid	2164.22	931.45	01.38	88%	92%

activation functions. MLP uses backpropagation for training the network. A simple MLP model is shown in Fig. 8. The steps involved in MLP learning is as follows:

- The forward propagation, where the data is propagated forward from the input layer to the output layer.
- The difference between the predicted and desired outcome which is term as error is calculated. The goal is to keep the error minimum.
- Backpropagate the error and update the model by determining its derivative with respect to each network weight.
- To learn the optimum weights, repeat the three processes listed above until a desired error is obtained or till the maximum generation is reached.

The activation function plays a very important role in a neural network, without which the model would be a linear regression model. All the non-linear transformations to the input are made possible by the activation function. Without an activation function, the simple addition of a hidden layer in the neural network would not produce any benefits, as the composition of linear functions would again yield only linear functions.

The performance of multiple neural networks with varying activation functions will be analysed in this paper. We also considered Deep Neural Network (DNN) which is an artificial neural network with deep layers. Deep layers means that the network contains numerous hidden layers for data processing and learning. Some of the activation functions used are hyperbolic tangent, identity, sigmoid, and softmax. Table 7 summarises the classification model performance when the hidden layer and output layer are given different activation functions. We have identified two observations from Table 7. First, DNN and MLP perform better than the others when a hyperbolic tangent is assigned to the hidden layer and a sigmoid function to the output layer. Second, the performance of MLP classification outperforms the DNN. Fig. 7 displays the receiver operating characteristic (ROC) curve for the MLP classification when various activation functions are assigned to the hidden and output layers. The ROC curve in Fig. 7(a), where the hyperbolic tangent and sigmoid activation function are applied to the hidden and output layers of the MLP classifier respectively, is closer to the left corner, signifying better performance. Therefore, we decide to use this activation function assignment for the following updated MLP classifier.

3.5. MLP-GA as proposed classifier

It is evident from the above discussion that we are not yet able to achieve a high classification performance. Now, we have improvised the work of a simple MLP classifier with the deployment of a GA. As shown

in Fig. 8, to find the appropriate weights for the neural network (NN), GA is deployed. The steps required in a MLP-GA algorithm are given below:

Step 1. Initial population

The process begins with a random initialization of a set of individuals called a population using binary strings. The individuals in MLP-GA matured step by step after defining the gene and then the chromosomes. In the paper, the size of the training input, which is the input neuron (N_I) is considered to 9 with three Hidden Neurons (N_H) and an Output neuron (N_O).

11 to 19 as shown in Fig. 8 represents the feature number 2, 5, 19, 21, 24, 35, 41, 53 and 67 respectively and Pi represents the input tuple. Eq. (4) provides the formula for the total weight (T_W), which is computed to be 30.

$$T_W = N_I \times N_H + N_H \times N_O \quad (4)$$

Eq. (5) gives the formula for the gene length (G_{length}). Considering a 16 bit binary number (B), the G_{length} is computed to be 480.

$$G_{length} = (B \times T_W) \quad (5)$$

A chromosome has two representations first one is the genotype and the second is the phenotype. In the paper, the genotype will be of 480 bits binary string.

Step 2. Construction of phenotype from the genotype

This step is the actual physical representation of the chromosome.

$$Y_i = \sum_{k=1}^{16} \gamma_{ik} 2^{-k} \quad (6)$$

where, γ_{ik} represents the k th bit for the i th weight, then using Eq. (6) we have:

$$X_i = Y_i \times \alpha_s + \beta_s \quad (7)$$

where, X_i represents the i th weight present in the solution with α_s and β_s denoting the scaling and the shifting factor respectively. As we are considering the weight value from $[-1, 1]$, we assign $\alpha_s=1$ and $\beta_s=0.5$.

The weights Q_{ji} and R_{kj} are computed in similar manner. Where Q_{ji} is the weight from the i th N_I to the j th N_H and R_{kj} is the weight from the j th N_H to the k th N_O .

Step 3. Calculating the outcome of hidden and the output layers

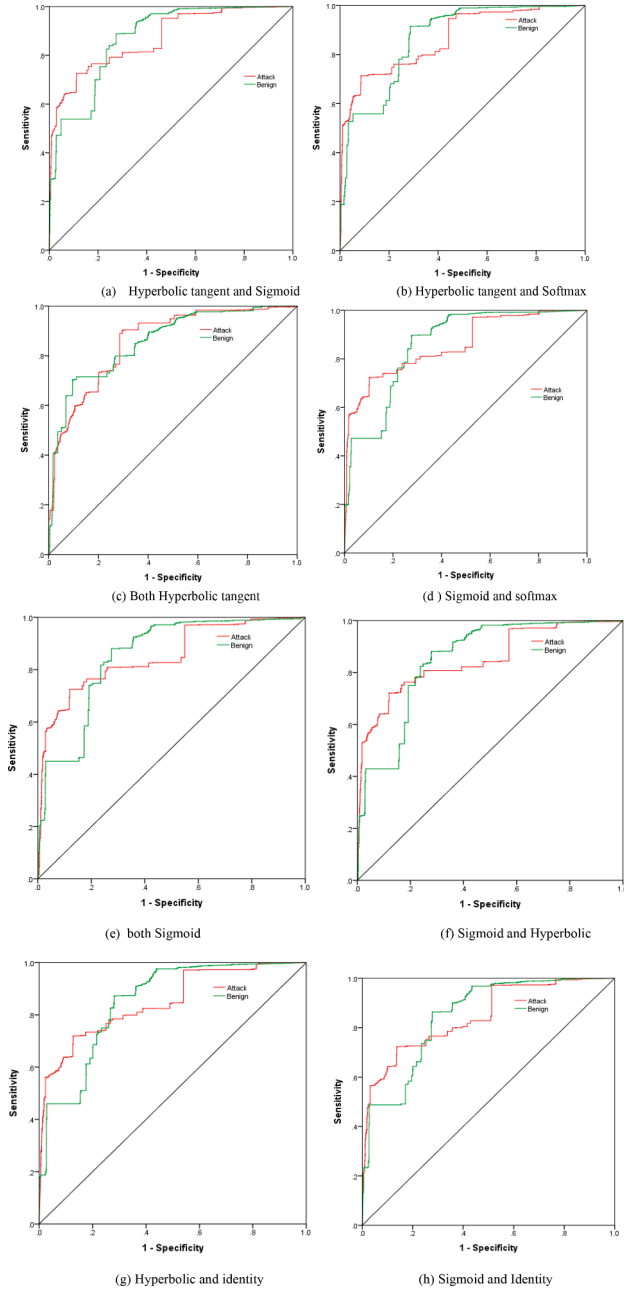


Fig. 7. ROC curve of the MLP classifier with various activation functions.

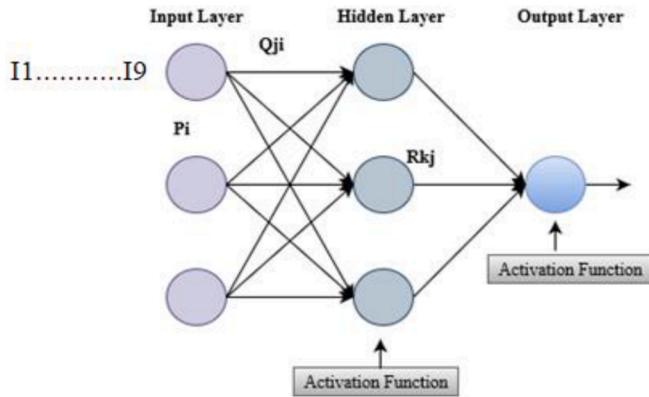


Fig. 8. Structure of the MLP network.

Suppose P represents the training sample from the training set and calculate the outputs of the hidden neurons as below:

$$\delta = \sum_{i,j} Q_{ji} \times P_i \quad (8)$$

$$Y_i = 2 \times \text{sigmoid}(\delta) - 1 \quad (9)$$

From Eq. (8) and hyperbolic tangent function we calculate Y_i using Eq. (9), which is the output of the j^{th} hidden neuron.

We now calculate the output from the output neurons;

$$\vartheta = \sum_{j,k} R_{kj} \times Y_i \quad (10)$$

$$\mu_k = \text{sigmoid}(\vartheta) \quad (11)$$

where, μ_k denotes the output of the k^{th} output neuron.

The change in error denoted by φ is updated using Eq. (11);

$$\varphi = 1/2 \sum_{k=0}^K (\rho_k - \mu_k)^2 \quad (12)$$

where the desired output is denoted by ρ_k . The computation proceeds till all the training sample are completed.

Step 4. Fitness calculation

The fitness function is calculated using the calculated error in Eq. (12) and number of training samples denoted by N.

$$F = \frac{N - \varphi}{N} \quad (13)$$

The above procedures from step number two are repeated for all the solutions of the population.

Step 5. Selection

Using Eq. (13), the strings with the greatest fitness value are selected. If the fitness value is greater than 0.99, then the computation is halted, as we have set the desirable fitness value to be 0.99. If the desired fitness value cannot be found, the algorithm terminates when it reaches the maximum population.

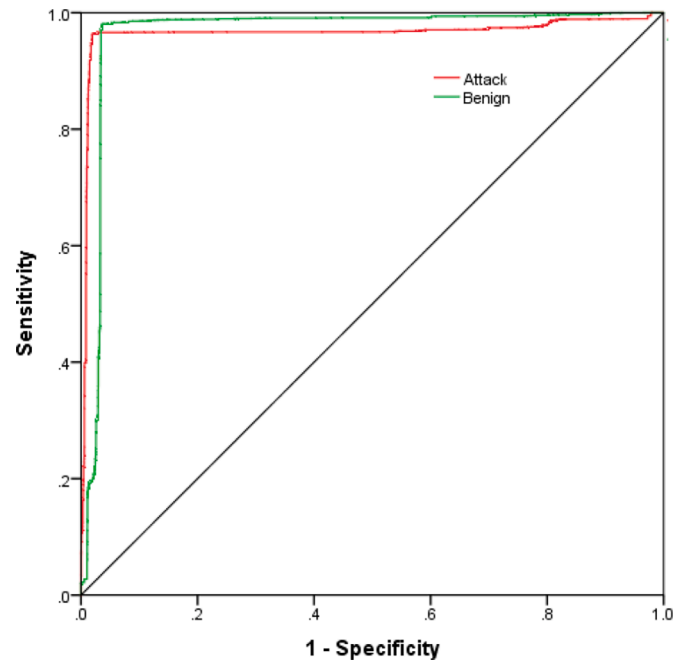


Fig. 9. MLP-GA performance with 10 features.

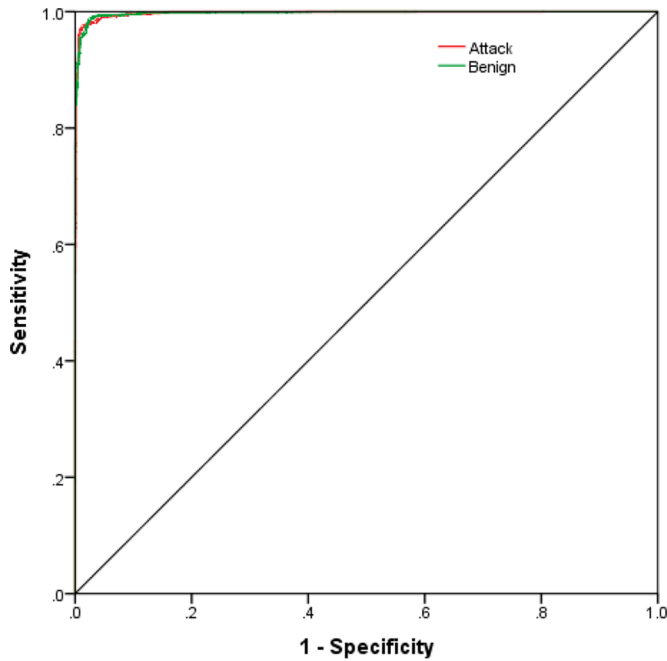


Fig. 10. MLP-GA performance with 9 features.

Table 8

Confusion Matrix of MLP-GA with 9 features.

Predicted Values	Actual Values	
	(TP)	(FP)
	14,005	5
	(FN)	(TN)
	175	815

Table 9

Confusion Matrix of MLP with 9 features.

Predicted Values	Actual Values	
	(TP)	(FP)
	13,055	75
	(FN)	(TN)
	1050	820

Step 6. Crossover

Crossover is a must to eliminate offspring that are identical to their parents. A uniform crossover technique is deployed by taking a subset from one parent and another subset from a different parent.

Step 7. Mutation

A scrambled mutation is performed by considering a subset of genes from the entire chromosome.

Step 8. Reproduction

After Steps 6 and 7, we get a modified population. The entire process, starting from Step 2, is repeated till we get a solution having the desired fitness value or the maximum number of generations have been reached.

4. Results and discussion

A ROC curve indicates the trade-off between sensitivity and specificity. The ROC curves of MLP-GA with two sets of features (Feature-10

Table 10

Performance parameter comparison between MLP and MLP-GA through multiple feature sets.

No. of Features	Performance	MLP	MLP-GA
Feature-78	Accuracy	88.44	89.4
	False Positive Rate	10.6	9.9
	Precision	95.6	97.5
	Recall	89	90
	F1-Score	92.2	93.6
	Error Rate	0.41	0.21
	Time taken (Sec)	32.3	45.8
	Accuracy	90.4	98.2
Feature-10	False Positive Rate	9.2	0.7
	Precision	98.8	99.3
	Recall	90.4	98.1
	F1-Score	94.4	98.7
	Error Rate	0.16	0.07
	Time taken (Sec)	2.33	2.55
	Accuracy	92.5	98.8
	False Positive Rate	8.3	0.6
Feature-9	Precision	99.4	99.9
	Recall	92.5	98.7
	F1-Score	95.8	99.3
	Error Rate	0.075	0.012
	Time taken (Sec)	1.54	1.52

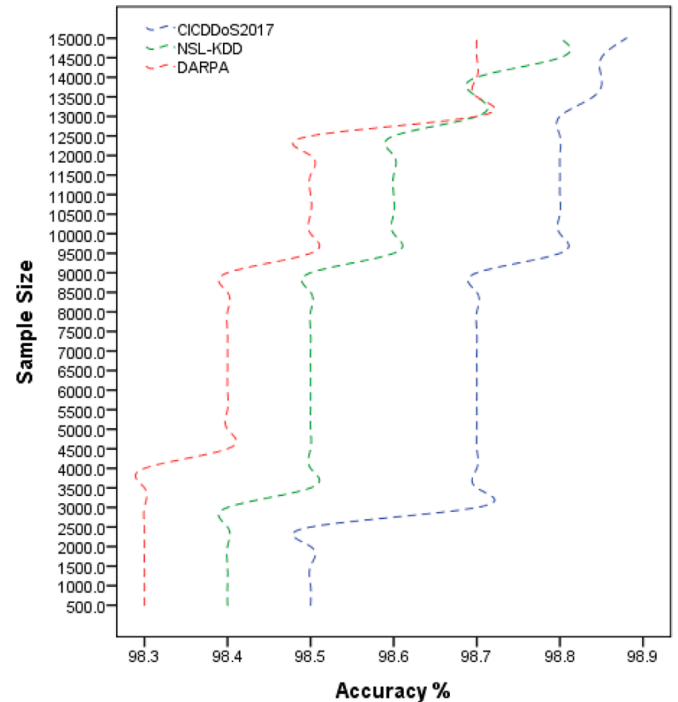


Fig. 11. Comparison of MLP-GA accuracy in three different Datasets.

and Feature-9) are shown in Figs. 9 and 10, respectively. In Figs. 9 and 10, it is found that the ROC curve of MLP-GA with the 9-Feature set is closer to the top left corner, indicating improved performance. The confusion matrix for MLP-GA and MLP using nine features is illustrated in Tables 8 and 9, respectively. We now compare the performance of MLP and MLP-GA for three different feature sets (Feature-78, Feature-10, and Feature-9), as shown in Table 10.

The formula for calculating Accuracy, False Positive Rate, Precision, Recall, F1-Score and Error rate is given by Eqs. (14)–(19), respectively. If a model's precision score is close to 100, it can be assumed that it correctly distinguishes between the attack and benign classes and didn't miss any true positives. A high F1 score denotes both high recall and great precision. Table 10 shows that MLP-GA outperforms MLP by having higher accuracy, precision, and F1 scores with a lower error rate.

Table 11

Comparison of the proposed method with the methods in the literature survey.

Paper	Method	Objective and Domain	Dataset	Performance Accuracy	Precision	Time Consumption	False Positive	Pros and Cons
[5]	1.Degree of DDoS attack to identify the DDoS attack. 2.KNN with ML	Detection SDN	Simulated Dataset	92%	99.3%	moderate	1.6%	Pre-identified features save time during feature selection, but their priorities are not mentioned, so important features are overlooked. Since KNN memorizes the entire dataset at the time of training, and at the time of prediction, it looks for the whole dataset to find the nearest neighbor, it is sometime known as lazy model. Hence, there is a need for large space requirement to store the entire training dataset
[6]	Trust based evaluation system based on trust elements	Detection VANET	VANET Dataset	95%	96%	high	n/a	It reduces bandwidth utilization and provides security for nodes; however, it faces several issues, such as a lack of essential evaluation data, the challenges of big data processing, and a lack of automation. One more limitation is that it suffers from scalability and optimization of node coverage in wireless sensor network.
[7]	Matching pursuit and wavelength with K mean Singular Value Decomposition (K-SVD) algorithm	Detection IDS (Intrusion Detection System)	CAIDA and BOUN Dataset	98.8%	99%	high	0.7%	It provides a matching pursuit algorithm to detect resource depletion and also generates dictionaries from the training dataset, but they are confined to low-density DDoS attack.
[8]	CRPS statistical mechanism and ES scheme	Detection IDS	DARPA99, MAWI and ICMPv6 traffic network datasets	94.13%	99.8%	n/a	0.18%	They combine the sensitivity of the exponential smoothing procedure with the ability of CRPS to distinguish between normal and abnormal features; however, CRPS is a single quantity, making it difficult to provide separate views on the prediction systems. The results indicate that the proposed method is appropriate only for major DDoS attacks. The model proposed is not accurate for medium and low traffic DDoS attack detection.
[9]	A hybrid method using Support Vector classifier with Random Forest	Detection SDN	Dataset is emulated using Mininet and Ryu Controller	98.8%	98.27%	moderate	2%	Does not contain any missing data; however, feature selection process is not automated, and all features are pre-defined. It is not applicable for large datasets. Since SVC model has a high computational cost, thereby increasing the computational cost of the overall ensemble approach.
[10]	AE for feature extraction and MLP for classification	Detection IDS	CICDDoS2019 Dataset	98.43%	97.91%	moderate	2.3%	Feature selection are automated, but there are high chances of eliminating important features with AE. A large portion of this work is limited to using a single dataset to determine how effective their proposal is.

(continued on next page)

Table 11 (continued)

Paper	Method	Objective and Domain	Dataset	Performance Accuracy	Precision	Time Consumption	False Positive	Pros and Cons
[11]	Characteristic Features based Neural Network	Detection and Mitigation IDS	Old dataset and new dataset Simulated Environment	92% with old and 98% with new datasets	96% with old dataset and 100% with new datasets	n/a	4%	It permits genuine packets to reach the server and prevents DDoS attacking packets from getting through. However, improper training or old patterns can result in poor detection. There is no discussion on how features are selected.
[12]	The EC tags legitimate traffic. The EC applications known as gates that reside at the network edge inspect for untagged or wrongly tagged packets and prioritize the tagged one.	Detection and Mitigation IDS	Local testbed with a linux-based implementation	n/a	n/a	high	n/a	The EC regulates the behavior and resource consumption of untagged and incorrectly tagged traffic. However, the packet-tagging system can cause a false alarm when dealing with large amounts of data streams from legitimate clients, and this will block the client by labeling them as attacks.
[13]	Combination of a Bayesian classifier with a Random Neural Network (RNN)	Detection IDS	DARPA 98 to 2000 datasets	n/a	n/a	n/a	n/a	It combines the powerful discriminating capacity and approximation properties of the RNN with both instantaneous values and statistical data of the incoming traffic. It is limited to flooding DoS attacks. They considered manual input feature selection.
[14]	Multilayer Perceptron as classification model using IP Flowdata	Detection IDS	Dataset generated using TFN2K tool	94%	96%	moderate	3%	They deployed a machine learning approach to avoid site-dependent thresholds that would demand great investment in terms of installation and customization. However, they are restricted to UDP-based DDoS attacks with limited work on feature selection.
Proposed method	Ensembled Feature selection technique with MLP-GA as classification model	Detection IDS	CICDDoS 2017, NSL-KDD, DARPA Datasets	98.8%	99.9%	1.52 S	0.6%	We presented a hybrid feature selection mechanism that provides effective, efficient, and relevant features with no redundancy. Three steps are given to provide a systematic method to feature selection. Selection of activation functions used in the MLP is illustrated. We introduced GA to MLP to solve the problem of gradient descent getting stuck in local minima and not changing the learning rate. GA provides global optimization. The proposed model provides better accuracy, high precision, F-1 score with reduced computation time and error rate.

As shown in Fig. 11, we yet again test the performance of MLP-GA using three distinct benchmark datasets, including CICDDoS 2017, NSL-KDD [31], and DARPA [32]. In all three benchmark datasets, MLP-GA is seen to be consistent for classification, with a marginally greater level of accuracy in the CICDDoS 2017 dataset.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (14)$$

$$False\ Positive\ Rate = 1 - specificity = \frac{FP}{FP + TN} \quad (15)$$

Algorithm 1**Final Feature Selection.**

```

Input: A set of 78 features from the original dataset
Output 1: A set of features whose value is greater than the mean priority value
Output 2: A set of 10 features
Output 3: A set of 9 Final Features
BEGIN:
FOR each Five Filters
Calculate the priority of each feature
Assign the features with highest priority value as Rank1 and lowest priority as Rank
78
Calculate the mean of the priority values obtained.
END FOR
FOR each Five Filters
IF (Priority value of each feature > Mean Priority value)
Select the feature
Else discard the Features
END IF
END FOR
Output1
FOR all the five filters
IF (a feature is present in all the five filter)
Select the feature
Else discard the feature
END IF
END FOR
Output2
FOR all the 10 features
Calculate the correlation using Pearson, Kendall and Spearman correlation
IF (0<Correlation value<1)
Select the features
Else Discard the feature
End IF
End FOR
Output3
END

```

$$Precision = \frac{TP}{TP + FP} \quad (16)$$

$$Recall = sensitivity = \frac{TP}{TP + FN} \quad (17)$$

$$F1\ Score = 2 * (precision * recall) / (precision + recall) \quad (18)$$

$$Error\ Rate = \frac{FP + FN}{TP + FP + FN + TN} \quad (19)$$

where, TP, TN, FP and FN stand for True Positive, True Negative, False Positive and False Negative respectively.

The comparison between the work that is being presented and the related works is shown in Table 11. It highlights the advantages of the proposed method among the related works in the survey. The feature selection limitations that were discovered in the majority of survey are now addressed.

5. Conclusion and future works

In the paper, one of our key goals is to increase feature selection's effectiveness. We reduced the dimension of the dataset by omitting features that are less effective in predicting the class category, features that are not related, and features that are redundant. We demonstrated this problem, which is not widely addressed by the community, through implementing multiple correlation methods. We introduced a dynamic GA-based MLP detection method against DDoS attacks, combined with an ensemble feature selection mechanism. We have observed that deploying all features available in the dataset reduces the accuracy of detection and increases the computation overhead. Similarly, omitting important features in the classification will generate irrelevant class categories. The hybrid feature selection, along with the correlation analysis cross-check, provides us with related features for classification.

The greatest challenge is being able to quickly and accurately detect DDoS attacks. We thus attain a 98.8% accuracy rate with a minimum false-positive rate of 0.6% and a computation time of 1.52 s using the GA-based MLP approach.

In the future, we intend to look into a more practical approach to achieving the rapid and accurate detection of various DDoS attack types. We'll concentrate more on putting the attack prevention mechanism into action.

CRedit authorship contribution statement

Usham Sanjota Chanu: Data curation, Writing – original draft, Conceptualization, Methodology, Software, Writing – review & editing, Formal analysis. **Khundrakpam Johnson Singh:** Visualization, Investigation, Software, Validation. **Yambem Jina Chanu:** Supervision, Resources.

Declaration of Competing Interest

There is no Conflict of Interest.

Data availability

Data will be made available on request.

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